Adaptive Spatial Tokenization Transformer for Salient Object Detection in Optical Remote Sensing Images

Lina Gao, Bing Liu, Ping Fu, *Member, IEEE*, and Mingzhu Xu, *Member, IEEE*

***Abstract*— Convolutional neural network (CNN)-based salient object detection (SOD) models have achieved promising performance in optical remote sensing images (ORSIs) in recent years. However, the restriction concerning the local sliding window operation of CNN has caused many existing CNN-based ORSI SOD models to still struggle with learning long-range relationships. To this end, a novel transformer framework is proposed for ORSI SOD, which is inspired by the powerful global dependency relationships of transformer networks. This is the first attempt to explore global and local details using transformer architecture for SOD in ORSIs. Concretely, we design an adaptive spatial tokenization transformer encoder to extract global–local features, which can accurately sparsify tokens for each input image and achieve competitive performance in ORSI SOD tasks. Then, a specific dense token aggregation decoder (DTAD) is proposed to generate saliency results, including three cascade decoders to integrate the global–local tokens and contextual dependencies. Extensive experiments indicate that the proposed model greatly surpasses 20 state-of-the-art (SOTA) SOD approaches on two standard ORSI SOD datasets under seven evaluation metrics. We also report comparison results to demonstrate the generalization capacity on the latest challenging ORSI datasets. In addition, we validate the contributions of different modules through a series of ablation analyses, especially the proposed adaptive spatial tokenization module (ASTM), which can halve the computational budget.**

***Index Terms*— Adaptive tokenization, optical remote sensing images (ORSIs), salient object detection (SOD), transformer.**

I. Introduction

SALIENT object detection (SOD) aims to segment the most salient pixelwise saliency region in a scene [1], [2], [3], [4], and it has been widely applied to various areas, including camouflaged detection [5], [6], [7], medical image segmentation [8], [9], [10], and hyperspectral image detection [11], [12]. Furthermore, it is noteworthy that the saliency is also broadly applicable to optical remote sensing images (ORSIs) over the past two decades, such as airport detection in ORSIs [13], [14], [15], ship detection in ORSIs [16], [17], instance or semantic segmentation in ORSIs [18], [19], and dehazing in ORSIs [20]. Recently, many remarkable methods for ORSI SOD have attracted increasing attention [15], [21], [22], [23], [24], [25], [26], [27], [28], which are generally divided into traditional models and convolutional neural network (CNN)-based models. For traditional ORSI SOD methods [15], [21], [22], the less representative handcrafted features (i.e., location cues, spatial structures, object boundaries or contrast between objects and backgrounds, and so on) limit their performance.

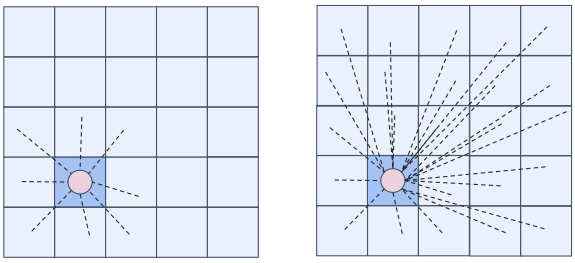


Fig. 1. Illustration of the receptive fields of different feature extraction methods. (a) CNN-based methods. (b) Transformer-based models. The saliency of the pixels is colored pink.

In contrast, many CNN-based ORSI SOD models have made significant progress in the recent decade years. The main reason is that CNN plays a key role in exploring more semantic cues for locating and segmenting salient regions. However, these CNN-based methods still suffer from the problem of learning long-range dependencies since local sliding window operations limit CNN-based methods to extracting global cues. Many studies have proved that global contrast [29], [30], [31] and global contexts [32], [33], [34], [35] are essential cues for SOD tasks. However, the receptive fields of CNN-based methods only cover the local region in Fig. 1(a), resulting in the loss of global details. To address this issue, some CNN-based ORSI SOD models obtain larger receptive fields through deeper layers or explore global cues via global pooling layers and nonlocal modules [26], [28], [36], [37]. However, these methods have difficulty obtaining long-range dependencies since the standard structures of the deep neural networks remain unchanged.

To this end, we hypothesize that a novel framework for exploring long-range dependencies is possible. The visual

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Lina Gao, Bing Liu, and Ping Fu are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: gaolina@hit.edu.cn; liubing66@hit.edu.cn; fuping@hit.edu.cn).

Mingzhu Xu is with the School of Software, Shandong University, Jinan 250101, China (e-mail: xumingzhu@sdu.edu.cn).

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saliency transformer (VST) [38] was the first technique to use the transformer and achieved promising performance in the SOD field. It proves that using a self-attention mechanism to replace neural networks has unlimited potential for SOD tasks. In Fig. 1(b), we can see that the transformer structure covers the entire image and models global long-range dependencies for each layer. To address the defect of CNN, the transformer architecture is considered extremely natural and necessary for exploring global cues and global contrast. Subsequently, different transformer variants for SOD have been developed and have achieved profound progress in fields such as RGB SOD [39], RGB-D/T SOD [40], [41], and video SOD (VSOD) [42].

Inspired by the transformer framework in NSI SOD, transformers are preferred for achieving better ORSI SOD performance. However, the characteristics of optical images are complex and changeable. The optical images may puzzle the background with the saliency region, which will lead to that the existing NSI SOD models exhibit insufficient performance for ORSI SOD areas. In addition, the unevenness of salient object distribution in ORSIs, such as salient objects off center and the scale of salient objects, is a too large or tiny and narrow salient object (NSO). At this time, no completely matched transformer variants are available for ORSI SOD tasks. Therefore, we design a specific transformer architecture with an adaptive spatial tokenization module (ASTM) to address these issues that are necessary, which can accurately locate the salient object location with fewer tokens. Different from existing transformer-based NSI SOD models, we can make the tokens adapt to the given ORSI data by learning to compute important regions with a spatial attention mechanism. It is difficult and time-consuming to divide every input into fixed patches and execute the transformer on patch sequence. Specifically, we capture the spatial weight from each token and obtain a new token sequence by multiplying input tokens. Then, long-range relationships are obtained for the selected tokens by the subsequent transformer modules.

In addition, fusing the tokens with various stages to predict saliency results is also an important issue for SOD. The incomplete or blurred boundaries of salient objects are particularly in ORSIs if all tokens are fused to predict saliency regions, which introduces some redundant tokens and weakens the impact of boundary details. Different from the VST [38] model that uses a reverse tokens-to-token (T2T) module to upsample tokens for obtaining saliency results, we capture more accurate global cues from sparse-scale tokens and then progressively integrate different stage tokens to generate high-quality saliency results via a specified designed DTAD. An important point here is that a structure loss function is used to supervise the three prediction maps at the pixel and object levels.

Our contributions are summarized as follows.

1) To the best of our knowledge, we are the first to design a novel dynamic spatial tokenization transformer for SOD in ORSIs, which can adapt tokens and significantly decrease the computational budget of the vision transformer in terms of total tokens.
2) Considering spatial inconsistency between different stages tokens, a DTAD is developed to progressively integrate the contextual dependencies. To refine the token with various scales and output high-quality saliency maps, we further optimize the model by using structure loss.
3) Comprehensive experimental results indicate the superiority of our proposed approach over ten NSI SOD methods and ten ORSI models on two ORSI SOD benchmarks. In addition, we conduct a comparison experiment on the latest challenging ORSI dataset to validate the generalization capacity of the proposed model. A series of ablation analyses are designed to reveal the contributions of each proposed component.

The remainder of the sections in this article can be organized as follows. An overview of SOD in NSIs, SOD in ORSIs, and visual transformer for SOD and three related works are provided in Section II. Our proposed model is detailed described in Section III. Furthermore, Section IV reports a comprehensive evaluation of our model and a series of ablation analyses. Finally, this article is concluded in Section V.

II. Related Works

A. Salient Object Detection in NSIs

Salient object detection in NSIs has ahieved significant progress over the past several decades, and the available methods traditional SOD models and deep CNN-based SOD models two types. In early years, traditional NSI SOD are usually focused on extracting handcrafted features, such as global region contrasts [29], [30], boundary backgrounds [43], color spaces [44], and center prior [45]. Chen et al. [30] proposed a regional contrast method, which simultaneously evaluates the differences and coherence of global contrast and spatial weighted to obtain high-quality saliency maps. Kim et al. [44] proposed a salient prediction model by applying a linear combination of colors in high-dimensional color space and utilizing a relative location as features.

Deep CNN-based NSI SOD methods have demonstrated the powerful expression abilities of features and have achieved significant performance improvement [34], [35], [46], [47], [48], [49], [50], [51]. For example, Deng et al. [46] designed a residual network to obtain complementary saliency cues and refine saliency maps. Zhao et al. [47] proposed an edge-guided model to address the complementarity between the edge cues of salient and saliency details, which can sufficiently refine the boundaries of salient objects. Pang et al. [48] constructed an interaction component to aggregate similar information in an encoder and then used a self-interaction component to extract multiscale cues for the decoder. This model achieves a significant performance improvement. Furthermore, Qin et al. [49] developed a nested U-structure for SOD for capturing more contextual cues with residual U-blocks. Xu et al. [51] designed a dynamic graph evolution model for learning multiple implicit cues from pairwise nodes. The above CNN-based SOD models for NSIs have achieved state-of-the-art (SOTA) performance, while these methods are not perfect for ORSI SOD.

B. Salient Object Detection in ORSIs

Different from NSI SOD, researchers have paid relatively little attention to ORSI SOD over the last few decades.

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Recently, research on ORSI SOD has been initiated. For instance, Zhang et al. [52] constructed a novel salient detection model for ORSIs, by using color cues content and obtaining full-resolution saliency maps. Furthermore, Zhang et al. [15] designed a dual CSA module to fuse low-level VOS cues with high-level KOS cues and proposed an SOT component to guide the energy functions learning. Liu et al. [22] built a circular feature map by using a parallel symmetric circle for SOD in ORSI. Zhang et al. [53] used a frequency domain analysis to efficiently extract attention and designed an adaptive threshold segmentation strategy to predict saliency results. However, the expression abilities of these abovementioned ORSI SOD models are limited by handcrafted features.

Therefore, inspired by the powerful expression abilities of CNNs, CNN-based SOD models have also achieved encouraging performance in ORSIs [23], [25], [28], [31], [54], [55], [56]. For example, Li et al. [36] proposed an end-to-end architecture, including a pyramid module and an encoder–decoder component for ORSI SOD. They also constructed the first ORSI dataset, which contains 800 images. Subsequently, Zhang et al. [25] designed an attention component to capture long-range dependencies details, a global context integration module, and a cascaded pyramid module for refining attention maps. In addition, a large-scale and challenging ORSI SOD dataset was constructed, including 2000 ORSIs with pixel-wise saliency labels. Li et al. [24] captured the coordination among neighboring features by an adjacent cue coordination network for SOD in ORSI. Cong et al. [56] first uses a parallel multiscale attention mechanism to design a relational inference model for ORSI SOD. A joint boundary and region model is proposed by Tu et al. [23] for generating more clearer boundary-aware ORSI saliency maps. They constructed a dataset containing 4199 ORSIs with different challenging scenes. Although these CNN-based ORSI SOD models have contributed significantly to performance improvement, they are limited by an inherent limitation of long-range semantic context relationships with CNNs.

C. Visual Transformer for Salient Object Detection

Transformer architectures can learn long-range dependency relationships with complete sequences in natural language processing (NLP) via self-attention mechanisms. Recently, inspired by the powerful abilities of transformer frameworks, many visual transformer models have been constructed for different vision areas, including object detection [57], [58], [59], SOD [38], [39], [40], [42], image recognition [60], [61], video understanding [62], [63], and so on, and have achieved promising performance. For SOD tasks, visual transformers also demonstrated their efficiency in NSI SOD [41], [64], [65]. For example, Liu et al. [38] first designed a unified model based on a visual transformer model for RGB and RGB-D SOD, and developed a multitask decoder to achieve saliency and boundary detection. Wang et al. [39] et al. proposed an "FCN + Transformer" framework with local and global perception capabilities for RGB SOD with four main modules.

To address the gap between the two modalities in RGB-D/T SOD, Liu et al. [40] proposed a dual-stream Swin transformer model with a spatial alignment and channel recalibration module to extract multimodal details and aggregate the intralevel features. A transformer-based model is proposed to capture implicit details and constructed a challenging RGBD CoSal1K dataset in [64]. This model has two class tokens for separately extracting intrasaliency and intersaliency information. In addition, Huang et al. [42] proposed a cross-reference transformer model with an integration of appearance and motion cues for VSOD. These transformer-based SOD models have achieved promising performance in NSIs though there is still a gap regarding the exploration of ORSI SOD with a transformer. Therefore, inspired by these transformer-based NSI SOD methods, we present the first attempt to design a transformer architecture for ORSI SOD in this article.

III. Proposed Model

In this section, we first elaborate on the overall architecture of our proposed adaptive tokenization transformer. As shown in Fig. 2, the overall model consists of an encoder with an ASTM and a DTAD. We describe the structure of the ASTM in detail in Section III-B, which is responsible for adjusting the number of tokens according to ORSI inputs. Then, the DTAD is designed to integrate saliency tokens and decode patches for predicting saliency results in Section III-C. We explain these components individually in the following.

A. Overview of the Proposed Architecture

The overall architecture of our proposed adaptive tokenization transformer model is illustrated in Fig. 2. Different from the fixed tokenization applied by the ViT [60] to the model input, we initially use the unfolding operation to progressively tokenize the given image to tokens. The unfolding operation is inspired by the success of the T2T module [66]. The scale and the number of salient objects are complex and changeable in optical images, thus to accurately and reasonably keep or drop tokens, our specified design a encoder with an adaptive tokenization module and typical transformer layers.

We assume that every ORSI data are $x \in \mathbb{R}^{h\times w\times c}$, each patch has a size of $k \times k$ with s overlap and p padding, and the length of the initial tokens T_0 is formulated as follows:

$$L = \left\lfloor \frac{h+2p-k}{k-s} + 1 \right\rfloor \times \left\lfloor \frac{w+2p-k}{k-s} + 1 \right\rfloor \tag{1}$$

where the $k - s$ operation is similar to the convolution operation, and the size of each patch is $k \times k \times c$. Then, T_0 is fed to the preceding N_1 transformer layers to extract its global–local details, obtaining new tokens T_1, which is calculated as

$$T_1 = \mathrm{MLP}(\mathrm{MSA}(T_0)) \tag{2}$$

where MLP and MSA are a multilayer perceptron with layer normalization and the multihead self-attention mechanism in the standard visual transformer [60], respectively. We empirically set the $N_1 = 14$ in this article.

Inspired by the positive performance of token learner for video recognition [67], the ASTM can adjust tokens in the next layer to further reduce the redundancy of numerous tokens, which is expressed as follows:

$$T_2 = \mathrm{AST}(T_1) \tag{3}$$

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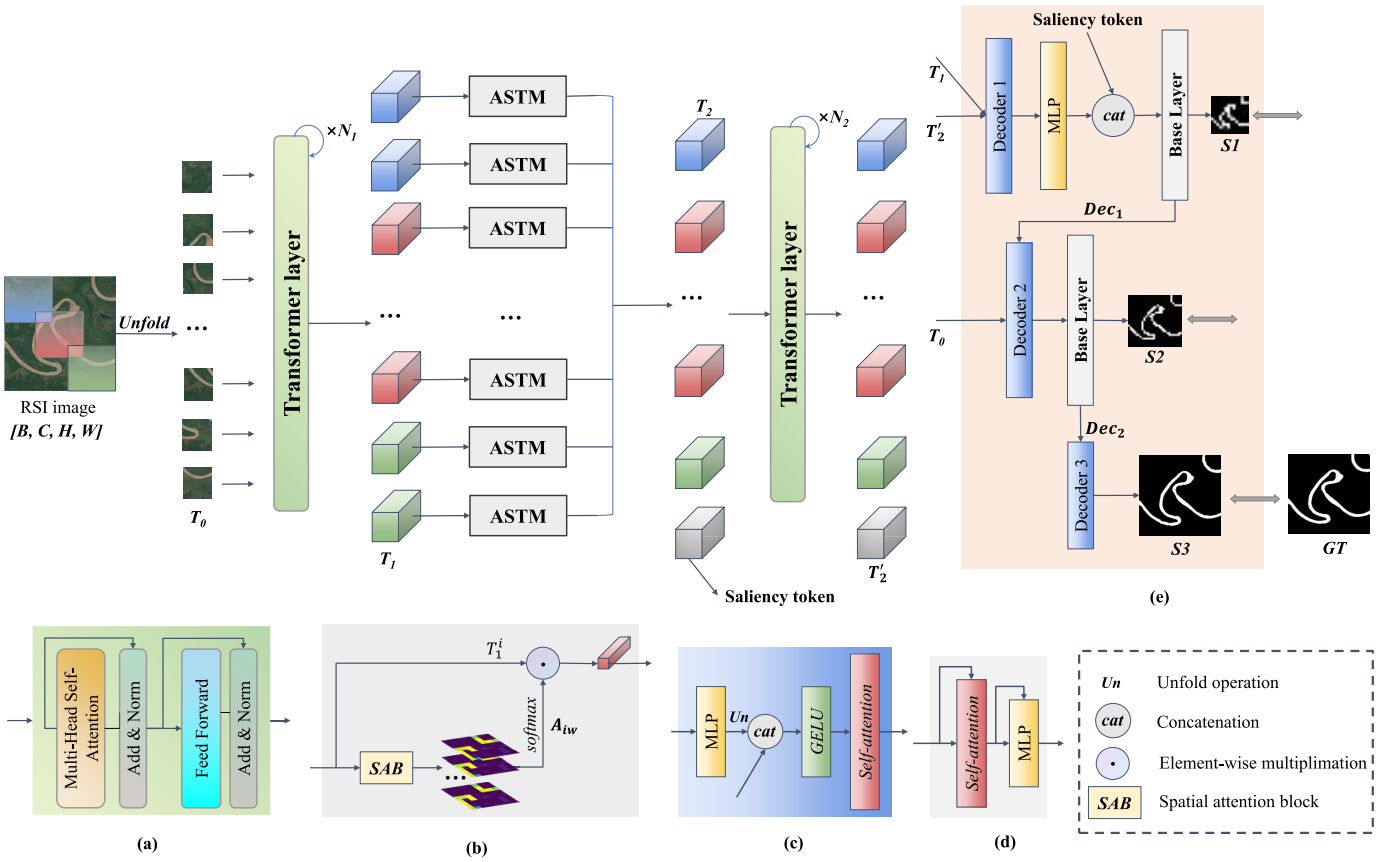


Fig. 2. Our overall architecture consists of two main components, including a backbone (encoder) with an ASTM and a DTAD. The input image is split into patches by an unfolding operation (a nonoverlapping split is shown here). (a) Transformer layer. (b) ASTM. (c) Decoder X. (d) Base layer. (e) DTAD. Best viewed in zoom.

where AST denotes the adaptive spatial tokenization operation, which is described further in Section III-B. Then, the saliency token is embedded in the selected token sequence T_2 to locate position cues and model long-range dependencies through N_2 transformer layers, which are defined as follows:

$$T_2' = \text{MLP}(\text{MSA}(T_2, s_t)) \tag{4}$$

where s_t is the saliency token and $N_2 = 8$. Finally, we use the proposed DTAD to decode tokens with saliency tokens and progressively integrate multiscale features to predict saliency regions.

B. Adaptive Spatial Tokenization Module

To select high-quality tokens from the complexity scenes in ORSIs adaptively, an ASTM is specified and designed for the tradeoff between the number and quality of tokens for the ORSI SOD. Specifically, the input image $X \in \mathbb{R}^{H\times W\times C}$ obtains L initial tokens T_1 by (2). Then, we formulate a token function, $SA_i = \text{AST}(T_1^i)$, where AST is a tokenizer function for learning an informative combination of pixels in T_1^i via the ASTM. The SAB is lighter and more efficient, and is implemented by some group convolutional layers with G_r groups to generate a spatial weight map. This way, we can dynamically adapt tokens by selecting spatial weights to significantly mitigate the side effects of the invalid tokens, which is computed as follows:

$$\text{SA}_i = \text{AST}_i(x) = \rho\left(T_1^i \odot A_{iw}\right) = \rho\left(T_1^i \odot \lambda\left(\alpha\left(T_1^i\right)\right)\right) \tag{5}$$

where $\alpha(T_1^i)$ represents elementwise spatial attention followed by a sigmoid function. In this article, our model sets the number of groups G_r to 16 due to the results in Section IV-E2. $A_{iw} \in \mathbb{R}^{H\times W\times C}$ is a spatial $H \times W \times 1$ weight map. $\lambda(\cdot)$ is a broadcasting function, and $\rho(\cdot)$ and $\odot$ denote spatial global pooling and elementwise multiplication, respectively.

From Fig. 2(b), the ASTM is embedded in each token, and then, dynamic learning is performed to select key pixels. We can summarize the process before the DTAD as an encoder of our model. Attention visualization examples with our proposed ASTM are illustrated in Fig. 3. We can clearly see that our module can adaptively adjust the number of tokens for ORSIs with various complexity levels, such as big salient objects (BSOs), multitiny salient objects, and NSOs. For example, we input a 224×224 ORSI with $k = 7$, $s = 4$, and $p = 2$. Considering the important local cues, a soft split of T2T module [66] is used to obtain 3136 tokens. However, considering that the computational cost increases with the processing of more tokens or layers, we use the tokens learned by the ASTM in the subsequent layers for saliency learning. The subsequent layers only process a few tokens (e.g., 196 instead of 3136). In particular, different locations of the ASTM in the overall architecture will yield different ORSI SOD performances. In this article, we insert the module after

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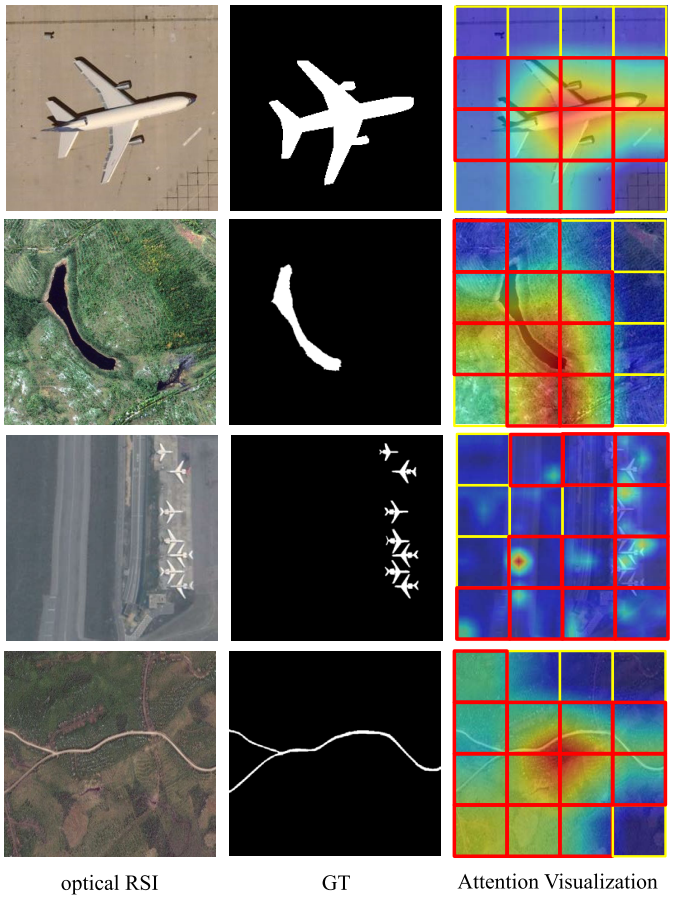


Fig. 3. Attention visualization examples with our proposed ASTM. The red block represents the selected intermediate tokens, and the yellow block is dropped tokens.

the second transformer layer (i.e., at the quarter point of the model); the detailed discussion is presented in Section IV-E.

C. Dense Token Aggregation Decoder

After obtaining the adaptive tokens, we extract more long-range context relationships by N_2 transformer layers and design a dense token aggregation decoder (DTAD) to aggregate these tokens for predicting high-quality saliency maps. To generate high-quality saliency results with a clear boundary and considering that a saliency map is consistent with the corresponding input optical RSI, we construct the DTAD with three subdecoders to gradually integrate the tokens from different stages.

As can be seen from Fig. 2(e), the DTAD consists of three cascade subdecoders. We first integrate T_2' with the saliency tokens s_t and T_1 to retain the structure details obtained from larger tokens and deeper channels. Considering that direct conversion is likely to lose significant details, we use a reverse unfolding operation [38] to unsample each token. Different from the dense prediction process of the VST [38], we use sparse-scale tokens to gradually expand and convert the original shape, which avoids the negative effect of redundant tokens. First, we decode T_1 and T_2' to obtain Dec_1. The scale of Dec_1 is $(H/8) \times (W/8) \times 1$, which is defined as

$$\text{Dec}_1 = \text{MSA}\big(U_n(\text{MLP}(T_1)), T_2'\big) \tag{6}$$

where the scale of Dec_1 is $(H/8) \times (W/8) \times 1$ and U_n denotes the reverse unfold operation of T2T to expand each token into multiple subpatches. Each token is defined as a $k \times k$ image patch with s overlapping and p padding. The computation process is the inverse to (1).

The original token T_0, which is refined by the complementary decoding map of T_2' and T_1, is calculated as

$$\text{Dec}_2 = \text{MSA}(U_n(\text{MLP}(T_0)), \text{Dec}_1) \tag{7}$$

where the scale of Dec_2 is $(H/4) \times (W/4) \times 1$.

We further decode the complementary map of the tokens from different stages and learn image-dependent saliency embeddings from the interaction tokens to obtain the saliency result, which is formulated as

$$S_3 = \text{MSA}(\text{MLP}(\text{Dec}_2)) \tag{8}$$

where S_3 is the saliency output and the scale of S_3 is the same as that of the original image that is $H \times W \times 1$. The two intersaliency maps S_i are defined by

$$S_i = \text{SA}(\text{MLP}(\text{Dec}_i)); \quad i = 1, 2 \tag{9}$$

where S_1 is $(H/8) \times (W/8) \times 1$ and S_2 is $(H/4) \times (W/4) \times 1$.

D. Loss Function

Finally, we jointly optimize the three saliency maps by applying the structure loss to impose the global and local restrictions, which is calculated as

$$\mathcal{L} = \mathcal{L}_s(G, S_3) + \eta\mathcal{L}_s(G, S_2) + \lambda\mathcal{L}_s(G, S_1) \tag{10}$$

where η and λ are hyperparameters for the tradeoff between the global and local details, and they are set to 0.6 and 0.5 according to the experimental results in Section IV-E1, respectively. G is the ground truth, and S_i, $i = 1, 2, 3$ represents the saliency result produced by (8) and (9); $\mathcal{L}_s$ is defined as follows:

$$\mathcal{L}_s = \mathcal{L}_{\text{IoU}}^w + \mathcal{L}_{\text{BCE}}^w \tag{11}$$

where $\mathcal{L}_{\text{IoU}}^w$ is the weighted intersection over union (IOU) loss function, which is calculated as follows:

$$\mathcal{L}_{\text{IoU}}^w = 1 - \frac{\sum_i^I \sum_j^J \left(s_{i,j} * g_{i,j}\right) * (\phi + 1)}{\sum_i^I \sum_j^J \left[\left(s_{i,j} + g_{i,j}\right) - \left(s_{i,j} * g_{i,j}\right)\right] * (\phi + 1)} \tag{12}$$

where $g_{i,j}$ and $s_{i,j}$ represent elements of the ground truth and the saliency prediction result; ϕ is defined as follows:

$$\phi = \left| \frac{\sum_i^I \sum_j^J g_{i,j}}{I * J} - g_{i,j} \right| \tag{13}$$

where $\phi \in 0, 1$. $\mathcal{L}_{\text{BCE}}^w$ is the weighted binary cross-entropy (BCE) loss function, which is calculated as

$$\mathcal{L}_{\text{BCE}}^w = \frac{\sum_i^I \sum_j^J \left[\left(1 - g_{i,j}\right) * \log\left(1 - s_{i,j}\right) - g_{i,j} * \log s_{i,j}\right]}{I * J}. \tag{14}$$

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IV. Experimental Results

In this section, extensive experiments are designed to demonstrate the superiority of our proposed model on three ORSI SOD datasets. First, we briefly introduce implementation details, the three ORSI SOD datasets, and the seven evaluation metrics. Quantitative results and qualitative analyses are reported to show the overall performance of our method compared with 20 SOTA models. We also validate the effectiveness of our proposed model in an ablation study. In addition, we conduct an evaluation experiment to indicate the generation ability of our model on the latest ORSI4199 SOD dataset and provide an analysis of some failure samples with challenging scenes.

A. Datasets

We comprehensively evaluate our model on three challenging ORSI SOD benchmarks, i.e., ORSSD [36], EORSSD [25], and ORSI4199 [23], to demonstrate the effectiveness of our method. These three ORSI SOD datasets contain 800, 2000, 4199 well-labeled ORSIs. The ORSSD is the first publicly available ORSI SOD dataset, which consists of many scenes selected from existing RSI datasets and Google Earth with items, such as playgrounds, islands, ships, and airplanes. Generally split it into 600 samples as the training set and the rest of 200 optical images as the testing set. The EORSSD expands the ORSSD dataset to 2000 images, making it the largest ORSI SOD dataset that has been recently released. It consists of 1400 training images and 600 testing samples. The ORSI4199 [23] is the latest dataset with 4199 images, including nine challenging scenes, i.e., a BSO, a complex scene (CS), a complex salient object (CSO), an incomplete salient object (ISO), a low-contrast scene (LCS), a multiple salient object (MSO), an NSO, an off-center (OC), and a small salient object (SSO). It has 2000 training samples and 2199 testing images.

B. Evaluation Metrics

Following existing representative literature, we report five widely employed performance metrics: the enhanced-alignment measure (E_ξ) [71], the structural measure (S_α) [72], the maximum F-measure (F_β) [73], the mean absolute error (MAE), and precision–recall (PR) curves. The PR curve calculates pairs of average precision and recall values with thresholds ranging from 0 to 255. In addition, we also provide two basic metrics, i.e., the IOU [74] and the overall accuracy (OA) [75].

To simultaneously consider the global means and local pixel matching, E_ξ is defined as follows:

$$E_\xi = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \xi_S(i,\ j) \tag{15}$$

where ξ_S reveals the saliency prediction and ground-truth relation after removing the means.

To evaluate the structural similarity between the saliency map and the corresponding ground truth, Fan et al. [72] calculated S_α using the following equation:

$$S_\alpha = \alpha \times S_{\text{object}} + (1 - \alpha) \times S_{\text{region}} \tag{16}$$

where α is empirically set to 0.5, as suggested in [72]. S_{object} and S_{region} refer to the salient object and region-aware structural similarities, respectively, which are defined in [72] in detail.

F_β is defined as the weighted harmonic mean of precision and recall

$$F_\beta = \frac{\left(1 + \beta^2\right) \times \text{precision} \times \text{recall}}{\beta^2 \times \text{precision} + \text{recall}}. \tag{17}$$

In [1], to emphasize precision over recall, β^2 is set to 0.3.

We also calculate the score of MAE; it can be defined as follows:

$$\text{MAE} = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} |\overline{S}(x, y) - \overline{G}(x, y)| \tag{18}$$

where $\overline{S}$ and $\overline{G}$ are normalized salient results and ground truth between 0 and 1, respectively.

The IOU [74] and OA [75] are the basic public evaluation metrics in computer vision. The IOU is calculated as

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{19}$$

where TP, FN, and FP represent the numbers of true positives, false negatives, and false positives, respectively. In this article, we report the OA measure, which is the average accuracy achieved on the whole testing set and is defined as follows:

$$\text{OA} = \frac{1}{N} \sum_{n=1}^{N} \text{OA}_n \tag{20}$$

where N represents all testing samples.

C. Experiment Settings

Our model is implemented on PyTorch [76] with the Adam optimizer [77], which loads the part weights of the publicly available $\text{T2T} - \text{ViT}_t - 14$ [66] model as the pretrained model. Following existing SOTA methods employed in [15], [21], [22], [25], [28], and [36], we utilize the same training datasets to guarantee a fair comparison. Specifically, we train our proposed model with 1400 original optical RSI images and ground-truth pairs from EORSSD [25] and the remaining samples from EORSSD (600) and ORSSD (200) as the test set. Our model training process takes 15 hours on one GPU (TITAN RTX). We set the learning rate, decay rate, batch size, and epochs to $1e-4$, 0.1, 16, and 200, respectively. For the latest proposed ORSI4199 [23] dataset, following Tu et al. [23], we use the same settings to retrain our proposed model and the comparison models (i.e., NLDF [32], PFAN [78], RAS [79], MJRBM [23], and HFANet [80]), and test their saliency performance on the test sets of ORSSD, EORSSD, and ORSI4199. We use multiple augmentation operations to augment the training data and resize the training samples to 224×224.

D. Comparison With the SOTA Methods

To conduct a fair comparison with the previously developed models, all models are trained on the EORSSD dataset

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TABLE I

QUANTITATIVE COMPARISON WITH TEN NSI SOD MODELS AND TEN ORSI SOD METHODS IN TERMS OF THE E_ξ, S_α, F_β, MAE, OA, AND IOU METRICS ON TWO BENCHMARK DATASETS (I.E., ORSSD AND EORSSD). WE ALSO REPORT THE FRAMES PER SECOND (fps) SCORES OF ALL MODELS. ↑ AND ↓ DENOTE THAT LARGER AND SMALLER VALUES OF THE CORRESPONDING METRIC ARE BETTER, RESPECTIVELY. THE "–" SYMBOL INDICATES THAT THE RESULTS ARE NOT AVAILABLE. WE MARK THE BEST RESULTS IN BOLDFACE

Type	Method	Publication	$FPS\uparrow$	ORSSD						EORSSD					
				$E_\xi\uparrow$	$S_\alpha\uparrow$	$F_\beta\uparrow$	$MAE\downarrow$	$OA\uparrow$	$IOU\uparrow$	$E_\xi\uparrow$	$S_\alpha\uparrow$	$F_\beta\uparrow$	$MAE\downarrow$	$OA\uparrow$	$IOU\uparrow$
NSI SOD Methods	RRWR	CVPR15 [43]	0.3	0.702	0.684	0.559	0.132	0.532	0.430	0.594	0.599	0.399	0.168	0.380	0.308
	HDCT	TIP16 [44]	7.0	0.650	0.620	0.526	0.131	0.508	0.332	0.638	0.597	0.541	0.109	0.533	0.320
	DSG	TIP17 [68]	0.6	0.734	0.720	0.624	0.104	0.627	0.476	0.659	0.642	0.523	0.125	0.518	0.378
	DSS	CVPR17 [69]	8.0	0.836	0.826	0.747	0.036	0.776	0.582	0.763	0.787	0.685	0.019	0.728	0.428
	R3Net	IJCAI18 [46]	7.0	0.868	0.814	0.746	0.040	0.694	0.620	0.829	0.818	0.750	0.017	0.787	0.524
	PoolNet	CVPR19 [70]	25.0	0.865	0.840	0.771	0.036	0.808	0.598	0.819	0.821	0.755	0.021	0.801	0.531
	EGNet	ICCV19 [47]	9.0	0.901	0.872	0.833	0.022	0.884	0.658	0.878	0.860	0.788	0.011	0.826	0.600
	MINet	CVPR20 [48]	12.0	0.945	0.904	0.876	0.014	0.892	0.782	0.935	0.904	0.834	0.009	0.834	0.742
	U2Net	PR20 [49]	30.0	0.939	0.916	0.874	0.017	0.875	0.777	0.937	0.920	0.873	0.008	0.880	0.746
	SUCA	TMM21 [50]	24.0	0.940	0.899	0.848	0.015	0.870	0.745	0.928	0.899	0.823	0.010	0.832	0.697
ORSI SOD Methods	VOS	GRSL18 [15]	-	0.598	0.537	0.417	0.215	0.292	0.197	0.651	0.508	0.360	0.210	0.244	0.157
	SMFF	IJRS19 [21]	-	0.774	0.531	0.486	0.185	0.401	0.186	0.740	0.541	0.574	0.143	0.505	0.211
	CMC	RS19 [22]	-	0.680	0.603	0.421	0.127	0.316	0.266	0.706	0.580	0.663	0.106	0.252	0.207
	LVNet	TGRS19 [36]	1.4	0.946	0.882	0.826	0.021	0.798	0.695	0.928	0.864	0.782	0.015	0.772	0.616
	DAFNet-R	TIP20 [25]	26.0	0.954	0.919	0.893	0.011	**0.925**	0.776	0.929	0.918	0.861	**0.006**	**0.891**	0.724
	SARNet	RS21 [54]	47.3	0.948	0.913	0.885	0.019	0.876	0.812	0.955	0.924	0.871	0.010	0.845	**0.793**
	MJRB	TGRS21 [23]	32.0	0.942	0.920	0.884	0.016	0.900	0.772	0.935	0.920	0.866	0.010	0.875	0.707
	RRNet	TGRS21 [56]	41.5	0.954	0.934	0.901	0.011	0.911	0.798	0.941	**0.927**	**0.878**	0.008	0.884	0.735
	ERPNet-R	TCyB22 [28]	12.0	0.960	0.935	0.904	0.011	0.891	0.810	0.937	0.925	0.874	0.008	0.873	0.743
	FSMINet	TGRS22 [55]	28.0	0.967	**0.936**	0.904	0.010	0.889	0.816	0.957	0.925	0.868	0.008	0.840	0.772
Ours	**Ours**	**-**	13.0	**0.970**	0.935	**0.906**	**0.009**	0.911	**0.826**	**0.958**	0.925	0.874	**0.006**	0.870	0.769

and obtained from their publicly available source. Specifically, we compare the proposed method with ten NSI SOD models, including RRWR [43], HDCT [44], DSG [68], DSS [69], R3Net [46], PoolNet [70], EGNet [47], MINet [48], U2Net [49], and SUCA [50], and ten optical RSI SOD methods that consist of VOS [15], SMFF [21], CMC [22], LVNet [36], DAFNet-R [25], SARNet [54], ERPNet-R [28], FSMINet [55], MJRBM [23], and RRNet [56]. We utilize the ResNet-50 version of DAFNet-R and ERPNet-R for a fair comparison as the parameters and FLOPs of ResNet-50 [81] are similar to those of the $\mathrm{T2T}-\mathrm{ViT}_t-14$ backbone.

1) Quantitative Results: In Table I, we report the quantitative results of the ten NSI SOD models and the ten ORSI SOD methods to quantitatively assess our model. We compute the scores of E_ξ, S_α, F_β, MAE, OA, and IOU metrics on the two ORSI SOD datasets. As can be seen from Table I, our proposed model achieves comprehensive ranks first on the two ORSI SOD testing benchmarks compared with the ten NSI SOD methods and ten ORSI SOD models, which further demonstrates the effectiveness of our model architecture. For example, comparing a DAFNet-R model that focuses on grappling with the limited long-range semantic relations of CNN-based SOD methods using an adaptive attention module, our model achieves outstanding performance and outperforms by averages of 2.25%, 1.65%, 1.3%, and 4.75% in terms of the E_ξ, S_α, F_β, and IOU metrics, respectively. The overall performance of our model remains steady despite the OA measure of DAFNet-R being slightly larger than that of our model. As the OA measure is insensitive to structural details, our adaptive transformer is highly capable of extracting global context information for ORSI SOD. Furthermore, our model is similar to the existing SOTA ORSI SOD models in terms of the MAE metric, and its absolute error is significantly improved over that of the NSI SOD methods. This may be attributed to the fact that our model can capture more discriminative features than existing NSI SOD models.

In addition, the PR curves are tested by our model and 20 SOTA methods on two ORSI SOD datasets, and these

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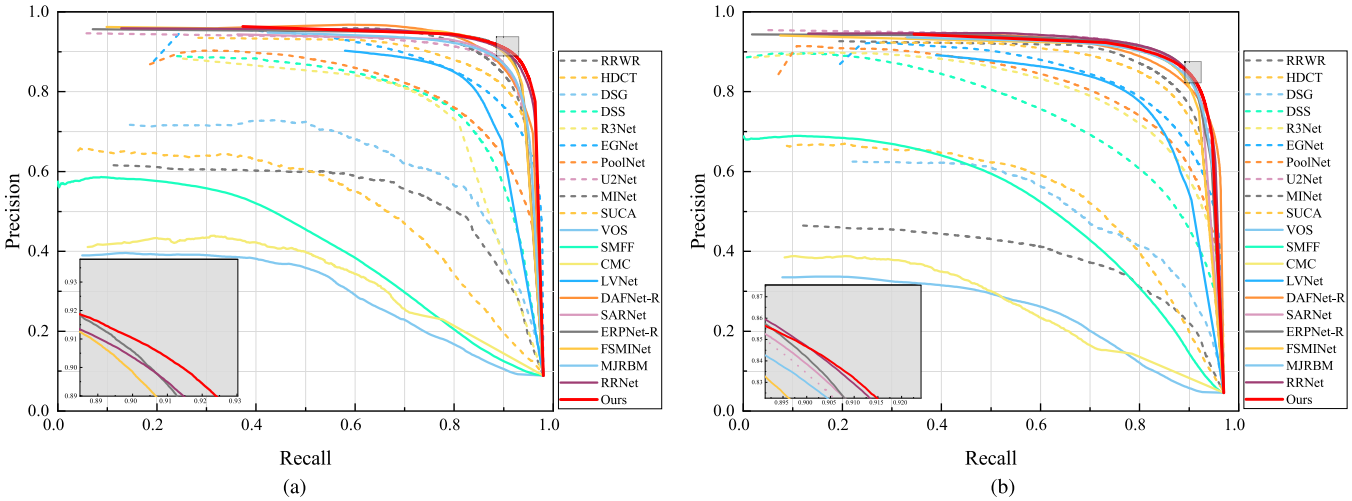


Fig. 4. PR curves tested by our model, ten NSI SOD models, and ten ORSI SOD methods on two datasets, i.e., (a) ORSSD and (b) EORSSD datasets. The dotted lines and continuous lines represent NSI SOD and ORSI SOD models, respectively. Zoom in for a better view.

curves are shown in Fig. 4. Specifically, the dotted lines and continuous lines represent the NSI SOD models and ORSI SOD methods, respectively. We further zoom in on two more intensive regions for clarity comparison. Our model achieves the highest precision rate among the tested models in most cases when the recall value is fixed. For example, as can be observed from Fig. 4(a), our model obtains the highest precision rate when the recall rate is 0.9 on the ORSSD dataset. In contrast, the FSMINet, RRNet, and ERPNet models obtain the lowest precision values compared with that of our model on the ORSSD dataset, which may be attributed to the fact that they cannot effectively explore global semantic features. The PR curves on EORSSD datasets shown in Fig. 4(b) demonstrate that our model is much higher than the curves of most existing SOTA methods. However, the curve of our proposed model is slightly lower than that of the RRNet model when the recall rate is 0.895 on the EORSSD datasets. It may be attributed to our model using the adaptive module, resulting in some beneficial cues being dropped. However, after weighing the pros and cons, we still use the ASTM to capture the long-range dependencies with saliency-related details.

2) Qualitative Comparison: Fig. 5 intuitively shows some saliency results selected from the EORSSD dataset with challenging scenes, such as salient objects off center, tiny and multisalient objects, NSO, and low contrast in specific circumstances. To maintain clear views, the qualitative results of our model and ten existing ORSI SOD models are given in this article. Compared with other models in Fig. 5, our model can completely predict salient objects in the most challenging scenes. For example, in the sixth row of Fig. 5, only our model and DAFNet-R, SARNet, and MJRB can predict four cars in a multitiny salient regions scene. The other seven models failed to locate the fourth car; the main reason is that these models are insufficient to explicitly model long-range context cues. In contrast, these failed models lose crucial global information due to the local sliding window operation of CNN. Furthermore, in LCSs, most existing SOTA models have difficulty predicting salient objects (eighth row in Fig. 5), except for our model and DAFNet-R. The main reason is that our model and DAFNet-R investigate long-range dependency relationships, which provide beneficial information and establish the locations of salient objects, differentiating them from the low-contrast background. However, compared with our model, DAFNet-R has blurred boundaries and introduces some background noise. From the discussion above, we can conclude that our model benefits from its powerful ability to excavate global semantic information from ORSIs.

E. Ablation Analysis

In this section, we first investigate the effects of the ASTM insertion position and three hyperparameters (i.e., η, λ, and the number of groups). A series of ablation studies are designed for analyzing the importance levels and the influence of our proposed two main modules, i.e., the ASTM, DTAD, and different loss functions. All variant models are retrained on the EORSSD datasets and reported four metrics (i.e., the E_ξ, S_α, F_β, and MAE metrics) on the two testing datasets.

1) Tradeoff Parameter Analysis: We first verify the influence of the tradeoff paramters i.e., η and λ in the loss function. As shown in Fig. 6, the performance curves obtained with different values of the two tradeoff parameters on two datasets, where four solid lines represent the four evaluation metrics, including E_ξ, S_α, and F_β, are as high as possible, and the MAE is as low as possible. The tradeoff parameter η in (10) is set to balance the impact of the initial and final saliency results. As shown in Fig. 6(a) and (b), the performance trends obtained with different settings of η are approximately the same for different datasets. Thus, we set this parameter to 0.6 to balance the weights between the losses of the global map and the local map. Similarly, the coefficient λ in (10) is also a balancing parameter, which controls the weights of the two-stage saliency map. We evaluate the SOD performance achieved balance parameter values of $\lambda = 0.0, 0.2, 0.4, 0.6, 0.8$, and 1.0 in terms of four evaluation metrics. From Fig. 6(c) and (d), we can observe that λ values that are too small ($\lambda \leqslant 0.4$) or too

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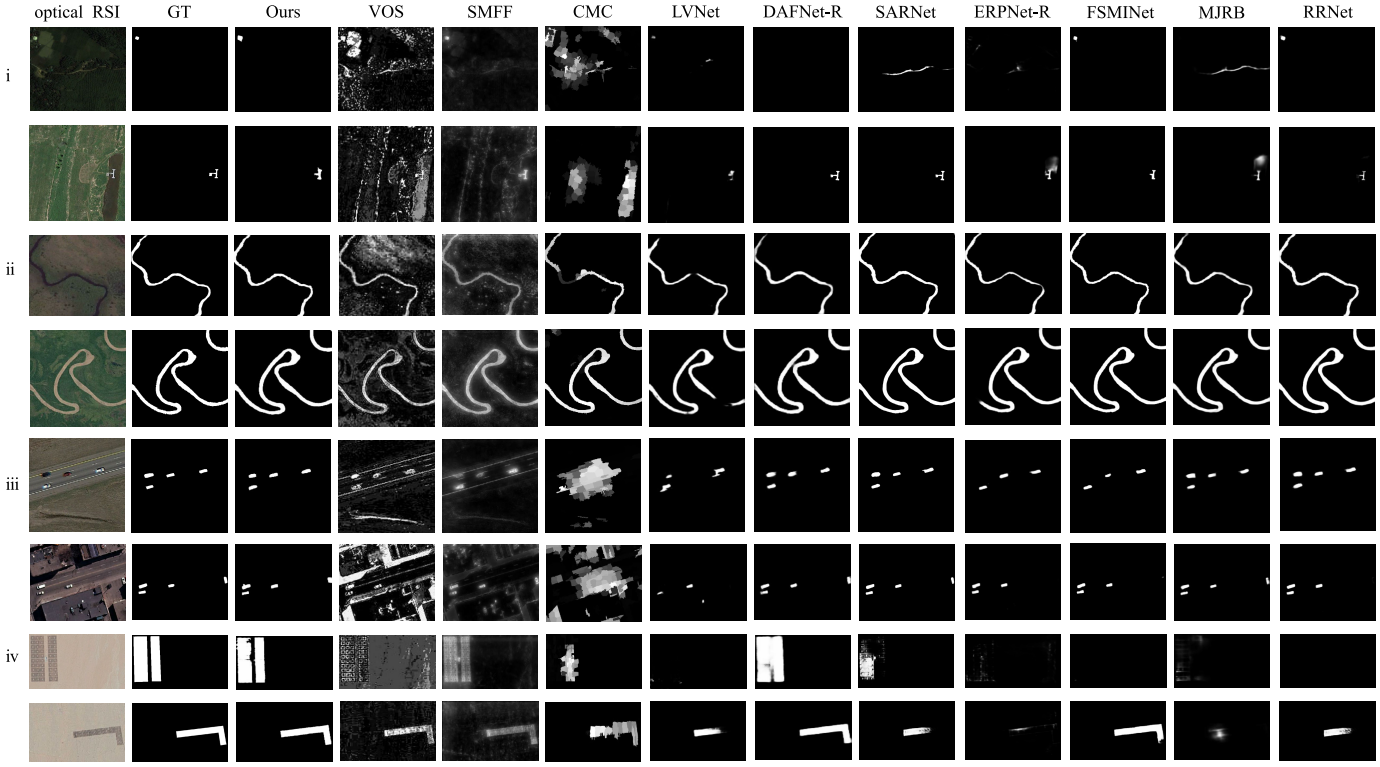


Fig. 5. Saliency results of our model on the EORSSD testing dataset compared with ten ORSI SOD approaches, including i: salient object off center; ii: NSO; iii: multitiny salient objects; and iv: low contrast in specific circumstances four challenging scenes. Best viewed in zoom.

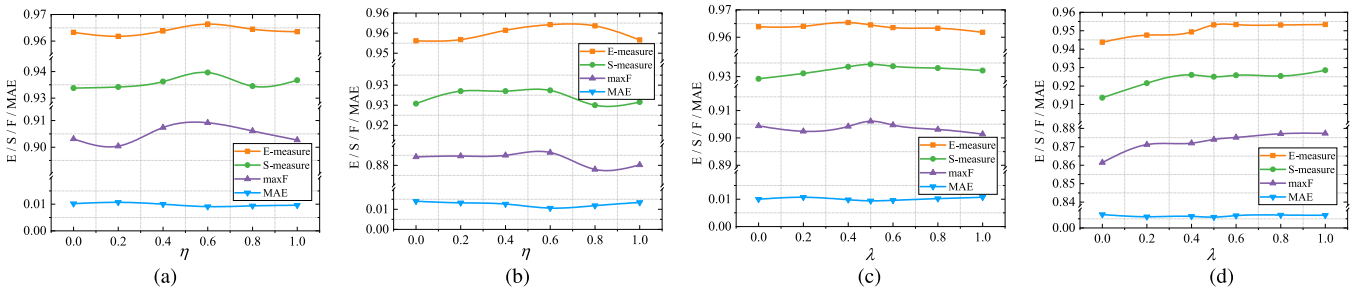


Fig. 6. Performance trends produced with different values of the two parameters, i.e., η and λ. The curves of E_ξ, S_α, F_β, and MAE on (a) and (c) ORSSD and (b) and (d) EORSSD.

TABLE II

COMPARISONS AMONG THE RESULTS OBTAINED WITH VARIOUS NUMBERS OF GROUPS IN TERMS OF THE E_ξ, S_α, F_β, AND MAE METRICS ON TWO ORSI SOD TESTING DATASETS. IN THIS ARTICLE, THE GROUP IS SET TO 16. WE USE BOLDFACE FOR THE BEST RESULTS

Variant	Group	ORSSD				EORSSD			
		$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
G8	G=8	0.952	0.927	0.898	0.012	0.930	0.918	0.862	0.007
G16	G=16	**0.970**	**0.935**	**0.906**	**0.009**	**0.958**	**0.925**	**0.874**	**0.006**
G32	G=32	0.955	0.934	0.905	0.010	0.930	0.921	0.867	0.007
G64	G=64	0.956	0.932	0.905	**0.009**	0.930	0.919	0.867	0.007

large λ value ($\lambda \geqslant 0.6$); all decrease the performance in terms of the four metrics. Thus, the parameter λ is set to 0.5 for a moderate balancing effect in the total loss function.

2) Influence of the Number of Groups: The group G is used to split the input and output channels at the SAB. Thus, to demonstrate the effect of different groups, we construct an ablation study setting this parameter to 8, 16, 32, and 64. The number of groups of these variants is different. "G16" sets the number of groups to 16 in Table II and Fig. 7. This version is our model in this article. From Table II, the "G16" model achieves comprehensive ranks first performance on two ORSI SOD datasets. In addition, the visualization results of "G8," "G32," and "G64" are incomplete, and their boundaries are exceptionally blurred in the seventh to ninth columns of Fig. 7.

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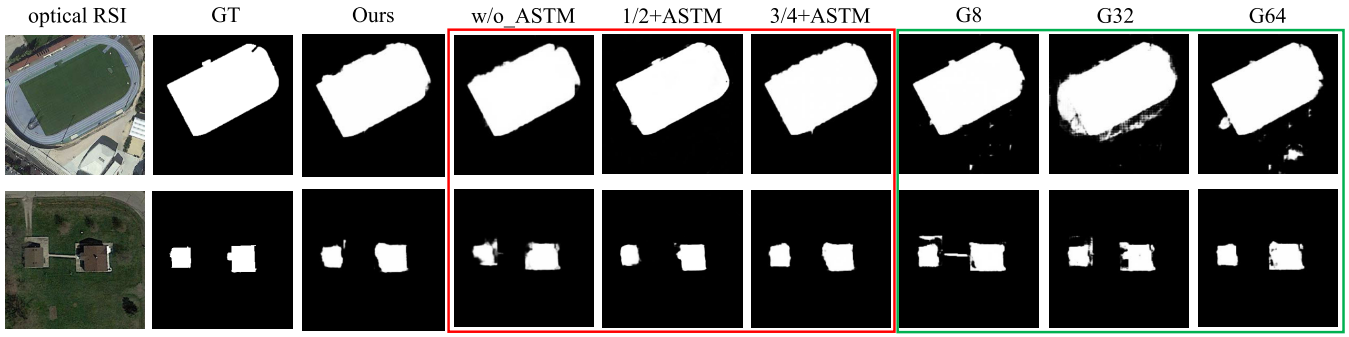


Fig. 7. Visualization results of the ablation experimentation concerning the ASTM on two challenging ORSI SOD datasets; there green and red blocks are the corresponding rows of Tables II and III, respectively. "Ours" is the entire model with the ASTM and 16 groups.

TABLE III

RESULTS OF AN ABLATION EXPERIMENT EVALUATING THE CONTRIBUTION OF THE ASTM, COMPARING IN TERMS OF THE E_ξ, S_α, F_β, AND MAE METRICS. "W/O_ASTM": THE ASTM IS REMOVED, AND THE NUMBER OF TOKENS IS FIXED. "1/4 + ASTM": WE INSERT THE ASTM AT THE QUARTER POINT OF THE MODEL, WHICH IS OUR MODEL. "1/2 + ASTM" AND "3/4 + ASTM" REPRESENT THAT THE ASTM IS INSERTED AT THE ONE-HALF AND THREE-QUARTERS POINTS OF THE BACKBONE, RESPECTIVELY. #PARAM (M) AND FLOPS (G) ARE OF THE ENTIRE NETWORK. WE USE BOLDFACE FOR THE BEST RESULTS

Variant	#Param (M)	FLOPs (G)	ORSSD				EORSSD			
			$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
w/o_ASTM	62.07	77.51	0.955	0.925	0.891	0.012	0.945	0.914	0.860	0.007
1/4 + ASTM	43.12	23.35	0.970	**0.935**	0.906	0.009	0.958	**0.925**	0.874	**0.006**
1/2 + ASTM	44.03	46.37	0.977	0.931	0.917	0.009	0.967	0.914	0.876	0.008
3/4 + ASTM	44.16	69.45	**0.979**	0.934	**0.921**	**0.008**	**0.973**	0.921	**0.885**	0.007

In theory, under the same computational conditions, we can set the number of groups as large as possible. However, using more groups will introduce more negative effects on exploring feature extraction results in a worse effect. Therefore, from these discussions, we can conclude that the group sets to 16 are much more worthwhile.

3) Effect of the Adaptive Spatial Tokenization Module: In this part, we first analysis the effect of the ASTM, as shown in the first row of Table III. The "w/o_ASTM" model removes the ASTM, while the initial parameters are the same as the proposed model ("1/4 + ASTM"). The "w/o_ASTM" model is also trained on the EORSSD training datasets. Furthermore, we also provide an efficiency analysis of different ASTM insertion positions to comprehensively compare the performance and computational costs. As can be observed from Table III, the overall score of the "w/o_ASTM" model is lower than that of the "1/4 + ASTM" model, decreasing by averages of 1.4%, 1.15%, and 1.45% in terms of E_ξ, S_α, and F_β, respectively. In addition, compared with our model, the "w/o_ASTM" model increases the computational cost, which is over three times higher than that of the "1/4 + ASTM" model, as the ASTM directly decides the quantity and quality of tokens. Furthermore, the third and fourth columns in Fig. 7 show the visualization results of our model and "w/o_ASTM" model. We can see that the 'w/o_ASTM' model is difficult to completely segment the salient regions from the background. Instead, our model can superior distinguish salient objects and backgrounds, which benefits from the explicit global contrast cues through our proposed ASTM. These results indicate that the ASTM can provide high-quality tokens for saliency prediction and adjust tokens to reduce computational costs.

4) Impact of the ASTM Insertion Location: We further analyze the influence of the ASTM insertion location in the model. Specifically, we insert the ASTM at the quarter point of the model, and this "1/4 + ASTM" model is our model in this article. "1/2 + ASTM" and "3/4 + ASTM" represent that the ASTM is inserted at the one-half and three-quarters points of the model, respectively. As illustrated in the second to fourth rows of Table III, compared with other variant models, the "3/4 + ASTM" model achieves the best performance on the two testing datasets. In addition, as can be seen from the third, fifth, and sixth columns of Fig. 7, the salient object boundary of "3/4 + ASTM" is more complete than that obtained with our model. However, the performance improvement of the "3/4 + ASTM" model requires computational cost and model complexity sacrifices. Therefore, we insert the ASTM at the quarter point of our model, which can significantly reduce the computational budget and achieve promising performance.

5) Importance of Dense Token Aggregation Decoder: To verify the contribution of the DTAD, the "w/o_DTAD" model uses direct conversion to upsampling tokens and element-wise summation to generate salient objects. As illustrated in Table IV, the "w/o_DTAD" model averages decrease by 1.25%, 1.25%, and 1.45% in terms of E_ξ, S_α, and F_β than our model on the two testing datasets, respectively. In addition, from Fig. 8, the visualization results intuitively illustrate the importance of our proposed DTAD for generating high-quality saliency maps. For example, the "w/o_DTAD" model failed to segment the three salient regions, and the boundaries are blurred from the first row of Fig. 8. In contrast, our model with the DTAD can completely segment the three islands, and the boundary of each island is quite sharp. This further validates

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TABLE IV

ABLATION EXPERIMENT ANALYZES EVALUATE THE IMPORTANCE OF THE DTAD. THE COMPARISON IS CONDUCTED IN TERMS OF THE E_ξ, S_α, F_β, AND MAE METRICS ON TWO ORSI SOD DATASETS. THE "W/O_DTAD" MODEL USES DIRECT CONVERSION TO UPSAMPLING TOKENS AND GENERATES SALIENT OBJECTS THROUGH A SIMPLE DECODER. WE USE BOLDFACE FOR THE BEST RESULTS

Variant	Components			ORSSD				EORSSD			
	$\mathcal{L}_{IoU}^{w}$	$\mathcal{L}_{BCE}^{w}$	DTAD	$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$E_\xi \uparrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
w/o_DTAD	✓	✓		0.958	0.922	0.892	0.012	0.945	0.913	0.859	0.008
w/o_BCE	✓		✓	0.957	0.920	0.884	0.014	0.945	0.908	0.849	0.008
w/o_IOU		✓	✓	0.958	0.922	0.883	0.012	0.944	0.904	0.845	0.007
Ours	✓	✓	✓	**0.970**	**0.935**	**0.906**	**0.009**	**0.958**	**0.925**	**0.874**	**0.006**

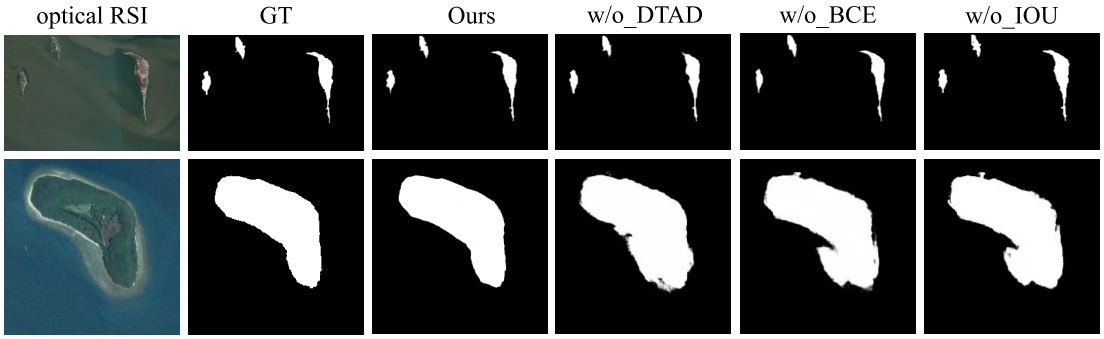


Fig. 8. Visualization results of ablation analyzes on two ORSI SOD testing sets, which corresponds to the row of Table IV.

that the DTAD can better integrate local and global semantic cues for obtaining saliency results.

6) Impacts of Different Loss Functions: We demonstrate the effects of different loss functions on performance improvement in this section. For comparison, we retrain two variants with different loss functions, including versions without the supervision of the BCE or IOU functions. Using the symbols "w/o_BCE" and "w/o_IOU" to represent the modified models in Table II, the performance comparison between "w/o_BCE" and "w/o_IOU" validates that the jointly optimized loss function can significantly improve the model performance. From Table II, the averages of our model improve by 1.3%, 1.6%, and 2.4% in terms of E_ξ, S_α, and F_β over those of the "w/o_BCE" model. This can be attributed to the fact that the simply weighted IOU loss function has a regression deficiency. Similarly, the "w/o_IOU" model average decreases by 1.3%, 1.7%, and 2.65% in terms of E_ξ, S_α, and F_β than our proposed model. In addition, the saliency results shown in the fifth and sixth columns of Fig. 8 show that our model can successfully suppress the background, while the "w/o_BCE" and "w/o_IOU" models introduce some black background in their salient object regions, and their boundaries are indistinct. The main reason is that the jointly optimized loss function with the BCE and IOU loss functions can encourage more consistent saliency details.

F. Generalization Analysis on a Latest Dataset

For a deep learning-based model, having a powerful generalization capability is important in real-world scenarios. Therefore, we provide an evaluation experiment to explore the generalization ability of our proposed model on the latest ORSI4199 [23] dataset. Table V reports the evaluation results.

TABLE V

PERFORMANCE OF DIFFERENT MODELS ON THE ORSI4199 DATASET [23]. THE COMPARISON MODELS AND OUR MODEL ARE TRAINED ON THE CHALLENGING ORSI4199 DATASET. LARGER VALUES OF $S_\alpha \uparrow$ ARE BETTER, AND SMALLER MAE VALUES ARE BETTER. THE "–" SYMBOL REPRESENTS THAT THE RESULTS ARE NOT AVAILABLE. WE USE BOLDFACE FOR THE BEST RESULTS

Method	Publication	ORSSD		EORSSD		ORS4199	
		$S_\alpha \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$MAE \downarrow$
NLDF	CVPR17 [32]	0.870	0.027	0.871	0.015	0.805	0.058
PFAN	CVPR19 [78]	0.885	0.021	0.885	0.013	0.837	0.049
RAS	TIP20 [79]	0.890	0.019	0.880	0.011	0.814	0.060
MJRBM	TGRS21 [23]	0.920	0.016	0.920	0.010	0.858	0.037
HFANet	TGRS22 [80]	0.938	0.009	**0.936**	0.007	0.876	0.033
Ours	-	**0.944**	**0.008**	0.933	**0.005**	**0.883**	**0.029**

Three NSI SOD methods, including NLDF [32], PFAN [78], and RAS [79], and two optical RSI SOD models, MJRBM [23] and HFANet [80]. These comparison models and our model are trained on the ORSI4199 dataset. In addition, each ORSI is labeled with attributes of challenging scenes in the ORSI4199 dataset. Thus, we report the results of two attribute-based comparison experiments, as shown in Tables VI and VII, which can reveal the advantages and disadvantages of different SOD models.

As can be seen from Table V, our model outperforms the previously developed SOTA SOD models on three testing datasets. Compared with RAS [79], our model achieves significant performance improvements and surpasses it by averages of 5.87% and 1.6% in terms of the S_α and MAE metrics, respectively. In addition, our model has stronger adaptability on the latest dataset than MJRBM and HFANet.

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TABLE VI

ATTRIBUTE-BASED PERFORMANCE ACHIEVED ON THE CHALLENGING ORSI4199 DATASET [23]. THESE MODELS ARE TRAINED AND TESTED ON THE ORSI4199 USING THE TRAINING AND TESTING DATASETS SPLIT PROPOSED IN [23]. THE LAST ROW CONTAINS THE AVERAGE STRUCTURAL SIMILARITY VALUES (S_α) FOR ALL TEST DATASETS WITH NINE CHALLENGING SCENES AS [23]. THE BEST RESULTS ARE MARKED IN BOLDFACE

Attr	NLDF CVPR17 [32]	PFAN CVPR19 [78]	RAS TIP20 [79]	MJRBM TGRS21 [23]	HFANet TGRS22 [80]	**Ours** - -
BSO	0.737	0.798	0.747	0.834	0.847	**0.887**
CS	0.740	0.802	0.765	0.845	0.848	**0.897**
CSO	0.707	0.771	0.731	0.800	0.816	**0.868**
ISO	0.650	0.728	0.660	0.787	0.810	**0.881**
LCS	0.631	0.676	0.682	0.733	0.743	**0.867**
MSO	0.730	0.774	0.723	0.818	0.808	**0.879**
NSO	0.672	0.750	0.736	0.839	0.863	**0.898**
OC	0.709	0.746	0.692	0.806	0.804	**0.865**
SSO	0.663	0.698	0.670	0.746	0.750	**0.857**
Avg	0.693	0.749	0.712	0.801	0.810	**0.878**

TABLE VII

ATTRIBUTE-BASED PERFORMANCE ON THE CHALLENGING ORSI4199 DATASET [23]. THESE MODELS ARE TRAINED AND TESTED ON ORSI4199 USING THE TRAINING AND TESTING DATASETS SPLIT PROPOSED IN [23]. THE LAST ROW CONTAINS THE AVERAGE MAX F-MEASURE (F_β) VALUES OBTAINED FOR ALL NINE CHALLENGING SCENES AS [23]. THE BEST RESULTS ARE MARKED IN BOLDFACE

Attr	NLDF CVPR17 [32]	PFAN CVPR19 [78]	RAS TIP20 [79]	MJRBM TGRS21 [23]	HFANet TGRS22 [80]	**Ours** - -
BSO	0.881	0.897	0.821	0.917	0.912	**0.918**
CS	0.826	0.862	0.813	0.896	0.889	**0.909**
CSO	0.821	0.857	0.811	**0.890**	0.880	**0.890**
ISO	0.860	0.874	0.818	0.894	0.904	**0.923**
LCS	0.712	0.762	0.736	0.792	**0.794**	0.785
MSO	0.774	0.806	0.814	**0.859**	0.846	0.857
NSO	0.680	0.844	0.690	0.875	0.877	**0.898**
OC	0.715	0.768	0.764	0.842	**0.845**	0.837
SSO	0.692	0.734	0.757	**0.809**	0.803	0.800
Avg	0.773	0.822	0.780	0.864	0.861	**0.869**

It benefits from the power of our model to explore the global context and enhance global–local dependency relationships. Tables VI and VII show the scores of S_α and F_β scores of our model and the five compared methods, respectively. Compared with the other models, our proposed method ranks first for nine attributes and averages in terms of the S_α metric. Although our proposed model ranks first on five of the nine attributes, our method also ranks first on average in Table VII. These results demonstrate that our model surpasses the existing methods in the most challenging conditions and has a powerful generalization ability.

G. Efficiency Evaluation

In Table I, we also evaluate the speeds of 20 compared models and our proposed method. In this part, we try our best to conduct a fair comparison with the previously developed models. All compared methods are generated by their released source codes. As some existing methods do not release source code, we only report relatively fair comparison results. The processing speeds are averages of those obtained on two testing datasets. As shown in Table I, the processing speed of our proposed model is slightly above average relative to 20 compared models. Transformer-based models are highly computational, while our model achieves a tradeoff between efficiency and effectiveness for the ORSI SOD task. Thanks to the ASTM, our proposed model can achieve a promising performance with a few parameters compared with the "w/o_ASTM" model in Table III. Furthermore, we find that our proposed model ("1/4_ASTM") holds a lowered memory usage. For example, when we input an optical image with $224 \times 224 \times 3$ and set batch size to 1, our model occupies only 333.70M memory. This may be attributed to the ASTM halving the computational budget.

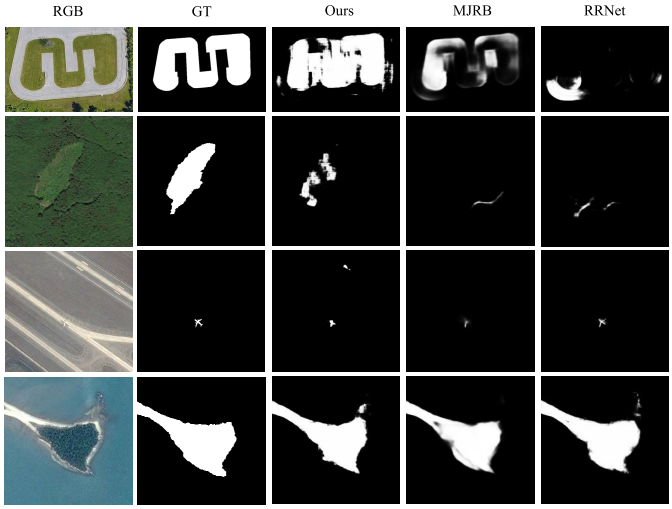


Fig. 9. Visualization results of failure cases on the EORSSD dataset with challenging scenes.

H. Analysis of Failure Samples

As previously mentioned, we aim to design a novel transformer framework that can adapt tokens and extract local and global semantic information for ORSI SOD. However, the proposed model still struggles with some limitations in a few challenging scenes. For instance, our model can not accurately predict the lawn with irregular geometry since the distributions of the ring road and salient regions appearance largely similarity, as shown in the first row of Fig. 9. The hilltop cannot be located in the second row since the salient regions are integrated with the background without noticeable boundaries, which confuses our model into believing that the surrounding background is also a saliency region. If the initialization information from the ASTM is wrong by beneficial information is wrong, our prediction results are undesirable.

Furthermore, in the third row, our model fails to predict tiny salient objects with low contrast, which may be due to the imbalance in training dataset distributions and the lack of extremely challenging scenes in EORSSD. The last scene with irregular geometries, contrast, and tiny salient objects

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is also challenging for existing SOTA models (i.e., MJRB and RRNet). The border disturbance of the salient region and background leads to all models failing to suppress the background, the highly similar in terms of appearance lead to three model fails to successfully suppress the background. However, compared with these CNN-based models, our model has an advantage regarding the power of its long-range dependency context cues. We will focus on designing a more effective ORSI SOD method to improve its discrimination ability for salient regions and disturbance regions in the future.

V. Conclusion

In this article, we investigate the characteristics of ORSIs and the long-range dependency extraction limitations of CNNs to propose an effective and efficient dynamic tokenization transformer framework for SOD in ORSIs. We first construct an ASTM as a flexible plugin and embed it in a transformer backbone for ORSI SOD, which can dynamically reduce tokens and retain the powerful capacity of the network to extract local and global details. In particular, the ASTM can be considered a plug-and-play module for reducing the redundancy of invalid tokens. Then, we propose a specific DTAD to progressively fuse the coarse tokens and various level details to generate saliency maps. Finally, a structure loss function is introduced to jointly supervise the three stages with respect to the local and global restrictions. Extensive experiments conducted on three ORSI datasets demonstrate that our proposed model outperforms 20 SOTA models. Each proposed component makes a significant contribution to the achieved performance improvement. Although our model has an advantage regarding the power of long-range context cues over CNN-based models, we plan to develop a more effective transformer-based ORSI SOD model for addressing some challenging scenes in the future.

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Lina Gao received the B.S. and M.Sc. degrees from the Northeast Forestry University, Harbin, China, in 2014 and 2017, respectively. She is currently pursuing the Ph.D. degree with the Department of Measurement and Control Engineering, School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin.

Her research interests include object tracking and salient object detection.



Bing Liu received the B.S., M.Sc., and Ph.D. degrees from the Harbin Institute of Technology (HIT), Harbin, China, in 2005, 2007, and 2012, respectively.

From 2015 to 2016, he was a Visiting Scholar with the University of Wisconsin–Madison, Madison, WI, USA. He is currently an Associate Professor with the School of Electronics and Information Engineering, HIT. His research interests include automatic test technologies, machine learning for image processing, and field-programmable gate array (FPGA)-based computing.



Ping Fu (Member, IEEE) received the B.S. degree from the Department of Radio Engineering, University of Science and Technology of China, Hefei, China, in 1989, and the M.Sc. degree in communication and electronic systems and the Ph.D. degree in measurement technology and instrumentation from the Harbin Institute of Technology (HIT), Harbin, China, in 1992 and 1999, respectively.

He is currently a Full Professor with the Department of Measurement and Control Engineering, School of Electronics and Information Engineering, HIT, where he is also the Vice-Director of the Automatic Test and Control Institute. His research interests include automatic test technologies, compressive sensing, machine learning, image processing, and field-programmable gate array (FPGA)-based computing.



Mingzhu Xu (Member, IEEE) received the B.S., M.Sc., and Ph.D. degrees from the Harbin Institute of Technology (HIT), Harbin, China, in 2013, 2015, and 2021, respectively.

He is currently an Assistant Professor with the School of Software, Shandong University, Jinan, China. His research interests include computer vision, multimedia computing and information retrieval, and explainable AI.

Dr. Xu is also an Invited Reviewer for prestigious journals, including IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS (TITS), Information Science, ACM Multimedia (MM), IEEE International Conference on Multimedia and Expo (ICME), and IEEE International conference on Pattern Recognition (ICPR).

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